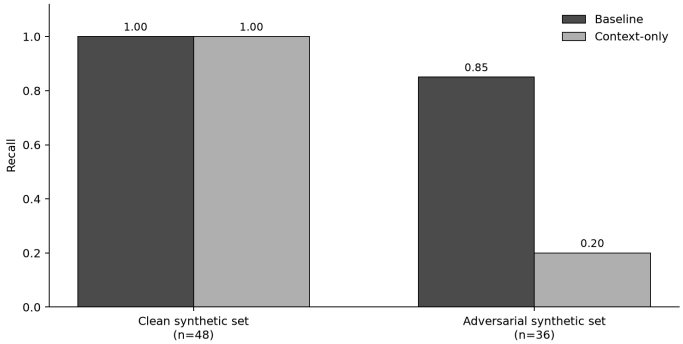
Screen PCI Redaction POC

Rishabh2503 12 October 2026 github.com/Rishabh2503/screen-pci-poc

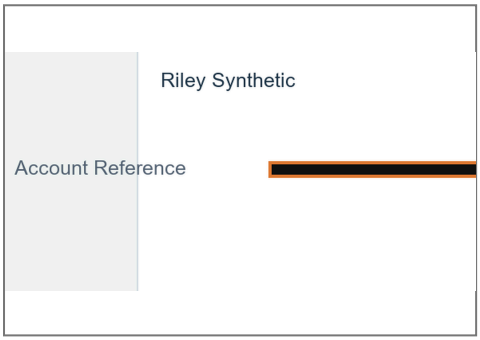
A context-aware redactor scored 1.00 recall on a clean synthetic set and 0.20 on an adversarial one (baseline: 0.85).

Final split: 15 images, 9 cards

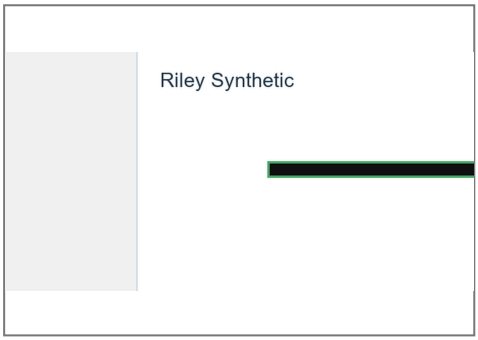
Recall: clean and adversarial synthetic sets



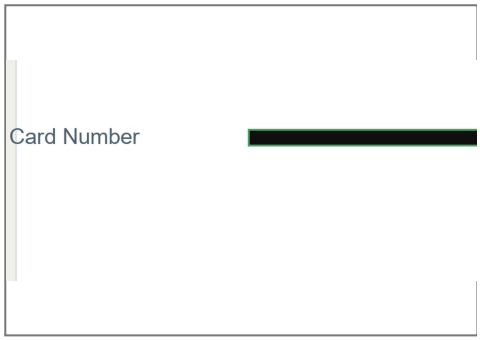
Sanitized failure examples



Baseline treated a Luhn-valid account reference as a card candidate.



Without label evidence, context-only rejected a true card candidate.



Cropped OCR text prevented the baseline from detecting the card.

Hybrid audit — final split

Recognition is measured on OCR output, before the masking decision.

Metric	Baseline	Context-only	Hybrid
Localization TP / FP / FN	8 / 4 / 1	2 / 1 / 7	9 / 5 / 0
Complete-number TP / FP / FN	8 / 4 / 1	8 / 4 / 1	8 / 4 / 1
Mean card-text coverage	88.0%	22.0%	94.9%
False redactions: decoy / ordinary-negative	4 / 0	1 / 0	4 / 1
Needs-review candidates: masked / total	0 / 0	0 / 4	6 / 6

Limits

All results are synthetic and template-limited. Hybrid mean card-box coverage is 94.9%, not full protection. Localization and exact complete-number recognition are different measures; these results do not establish real-world PCI protection or compliance.

What changed and what I learned

Adding label context removed clean-set false positives without changing recall on that set.

Adversarial layouts and labels reduced context-only recall to 0.20.

The hybrid policy restored localization and coverage, but also redacted ordinary-negative text.

This is a controlled synthetic experiment, not a validated data-protection system.